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To the Editors-in-Chief and Editorial Board,
Journal of Machine Learning Research (JMLR)

Subject: Submission of manuscript for JMLR (Regular Article)

Dear Editors,

I am writing to submit my manuscript entitled “**TACT: Trust-Anchored Confidence Tempering for Self-Consistency Voting in Large Language Models**” for publication as a regular article in the *Journal of Machine Learning Research*.

The paper addresses a structural gap in confidence-weighted self-consistency: every published weighting scheme is monotone *increasing* in confidence, so a channel whose confidence is anti-correlated with correctness poisons the vote, and binary calibration gates survive that inversion only by discarding discriminative signal. TACT derives the vote exponent from the measured, *signed*, within-item discrimination of the channel — a pooled van Elteren Somers’ D with an item-clustered standard error, positive-part James–Stein shrinkage, and a Bayes-discriminant link — so the exponent is derived rather than grid-searched, and the construction has exact endpoints: inside the shrinkage dead zone the vote is bit-identical to plain self-consistency.

I should be direct with the editors about what the paper concludes, because it is not a straightforward win. Alongside the method, the manuscript reports its own negative measurements: a dev-picked *signed* grid turns out to be a far stronger baseline than any published protocol; the stratum on which *any* label-free aggregation method can act measures 2.5–7.5% of items across five substrates in two domains; on both real-trace substrates an in-pool oracle cannot clear a pre-registered endpoint; and the label-free arm’s abstention on one substrate turns out to be structural rather than a reading of the channel — a defect found deliberately, by planting a channel of known strength and watching the method fail to act on it. The negative results are the contribution as much as the estimator is, and the paper is submitted on that basis rather than as a claim of improved accuracy.

As required by the JMLR submission guidelines, I provide the following.

- 1. Prior publications with significant overlap.** None. No part of this manuscript has been published or is under review elsewhere, and it does not extend a previous conference paper. For completeness: Section 2 cites an earlier unpublished negative result by the same author (a redundancy-discounted voting system with value-of-information stopping) as the work whose failure motivated this problem. No content, results or text from it appear here; the citation exists only to be honest about provenance, and can be removed if the editor prefers.
- 2. Co-author consent.** The manuscript is sole-authored. The single author consents to review.
- 3. Conflicts of interest and funding.** None. The research was conducted fully independently, received no external funding, and the author has no financial interest or personal relationship that could have influenced the work. The frozen model used in the real-trace campaigns was accessed as an ordinary paying API customer, with no relationship to its vendor.

4. **Five keywords.** Large language models; self-consistency; confidence calibration; label-free estimation; rank statistics.
5. **Suggested action editors.** All are current members of the JMLR editorial board. The author has no prior collaboration, co-authorship, shared affiliation or personal relationship with any of them.
 - **Kai-Wei Chang**, UCLA — large language models and trustworthy NLP; the setting is inference-time aggregation over LLM reasoning traces.
 - **Chris Oates**, Newcastle University — uncertainty quantification; the contribution is a signed, uncertainty-aware estimator with an explicit abstention region.
 - **Sivaraman Balakrishnan**, Carnegie Mellon University — theoretical statistics; the estimator is a van Elteren-pooled rank statistic with an exact tie-corrected null variance and a jackknife.
 - **Benjamin Guedj**, Inria and UCL — computational statistics; the Bayes-discriminant link and the attenuation identity under class-conditional noise most need an adversarial reading.
 - **Maya Gupta**, University of Washington — ensembles and safety; the paper’s recommendation that abstention is the correct default is an ensemble-safety argument as much as an accuracy one.
6. **Suggested reviewers.** Each is an author of work the manuscript cites and engages with directly, so each can check the comparisons rather than only the prose. The author has no conflict of interest with any of them.
 - **Dr. Xuezhi Wang**, Google DeepMind — originator of the self-consistency mechanism. The paper’s zero-trust anchor is a claim of *bit-identical* reduction to self-consistency, tie-breaks included, and should be checked by its author.
 - **Dr. Sebastian Farquhar**, University of Oxford — semantic uncertainty. The manuscript names semantic-equivalence clustering as the untried remedy for its sharpest unguarded failure mode, a paraphrased wrong majority that lexical deduplication cannot see.
 - **Prof. Edgar Dobriban**, University of Pennsylvania — rank-calibration. His work reaches the discrimination-over-calibration conclusion independently and by different means, so he is well placed to judge whether the signed extension is a real advance.
 - **Amir Taubenfeld**, Google Research — CISC and the Within-Question Discrimination metric, the closest baseline. The paper claims an exact reduction to CISC-power under a log-value feature map, and argues that CISC’s tuned temperature already interpolates dev-calibrated self-consistency and CISC.
 - **Fabio Parisi** — label-free combination of predictors. The identifiability boundary in the label-free variant is presented as the single-channel analogue of the two-root ambiguity in his work.

All code, cached traces, and the JSON artifact behind every table and figure accompany the submission as supplementary material. A script cross-checks all 42 quoted figures against those artifacts, and a 102-test suite pins every proposition in the paper to an executable counterpart.

Thank you for your time and consideration.

Sincerely,

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